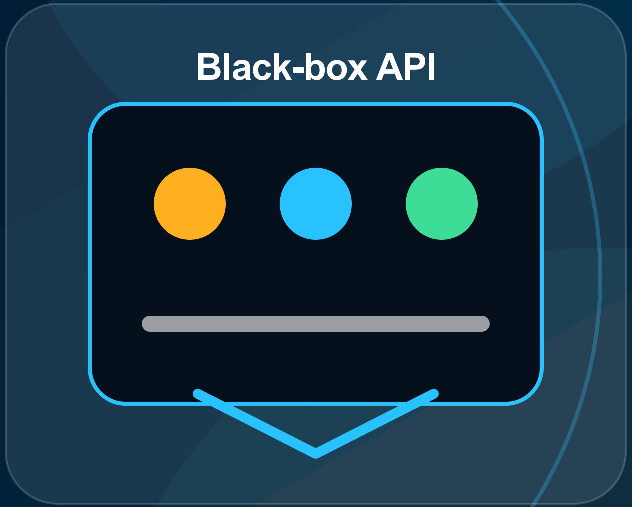
LLM API INTEGRITY AUDIT

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs

Why software tests fail and hardware attestation wins



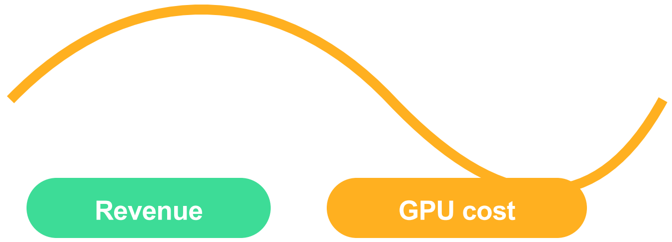
Commercial providers can covertly serve cheaper substitutes behind black-box APIs; users need verifiable inference, not probabilistic guesses.

Reference: arxiv.org/abs/2504.04715

The Model Substitution Problem: Economic Incentives Drive Covert Swaps

Economic incentive

Advertised premium model pricing meets GPU cost pressure. In a black-box API, users cannot directly verify which weights served each response.



Margin incentive

Four covert substitution classes

1. **Smaller model**
   e.g., 8B instead of 70B
2. **Quantized variant**
   FP8/INT8 instead of FP16
3. **Fine-tuned / updated**
   behaviorally close but not specified
4. **Different family**
   entirely different architecture

Hypothesis test: H_0: P_actual(y|x)=P_spec(y|x)
H_1: P_actual(y|x)≠P_spec(y|x)

EMPIRICAL BENCHMARK EVIDENCE

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Original vs. FP8-quantized benchmark scores (mean ± error, from source brief)

Model family	MMLU Orig	MMLU FP8	GSM8K Orig	GSM8K FP8	MATH Orig	MATH FP8	GPQA Orig	GPQA FP8
Llama-3-8B-Instruct	62.69 ± 0.18	62.43 ± 0.26	61.14 ± 3.47	60.90 ± 4.10	20.65 ± 3.43	14.91 ± 2.49	22.62 ± 0.22	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	77.88 ± 0.13	88.06 ± 1.44	87.35 ± 1.24	35.69 ± 1.33	35.75 ± 1.16	29.60 ± 0.51	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	71.92 ± 0.11	81.80 ± 1.35	79.41 ± 1.14	33.41 ± 0.28	32.53 ± 0.34	28.64 ± 2.97	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	81.98 ± 0.08	86.72 ± 1.00	86.82 ± 0.97	37.39 ± 1.41	37.67 ± 1.39	29.93 ± 2.71	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	58.77 ± 0.13	35.90 ± 4.54	32.20 ± 4.02	8.94 ± 1.24	7.68 ± 1.23	21.60 ± 0.17	22.72 ± 0.19

Interpretation: FP8 often stays within original error bars on MMLU and GSM8K; even when drops appear, benchmark variance and task sensitivity make black-box detection fragile.

SAMPLING TEMPERATURE STRESS TEST

Higher Temperature Amplifies Performance Gaps but Increases Noise

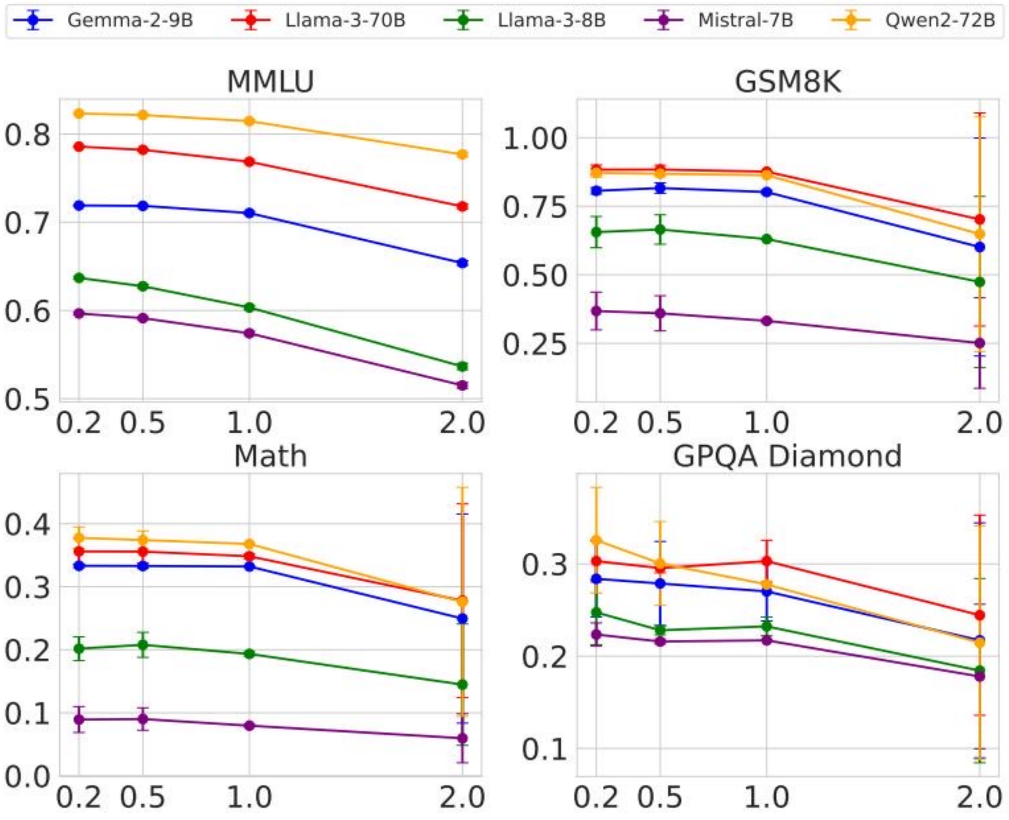


Figure 1: benchmark accuracy vs. temperature for five models across four tasks.

What changes at high temperature?

- **Performance degrades across the board**
  temperature 2.0 lowers benchmark accuracy for original and substitute models
- **Signal becomes noisy**
  sampling variability obscures whether the source is substitution or stochastic decoding
- **Detection becomes brittle**
  gap size depends on task, model family, and decoding setting

More samples ≠ clean attribution

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Figure 5: same Llama-3-8B model produces varying log probabilities across frameworks and hardware.

Reference match assumption

Auditor compares token-level log probabilities against a local reference model.

Production reality

Transformers/vLLM versions and A100/H100 hardware change the measured log-probabilities.

Resulting ambiguity

Infrastructure drift can look like model substitution; substitution can hide inside normal drift.

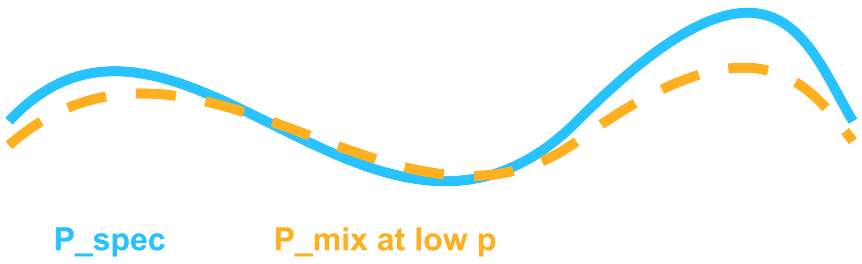
ADAPTIVE ADVERSARIAL ROUTING

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

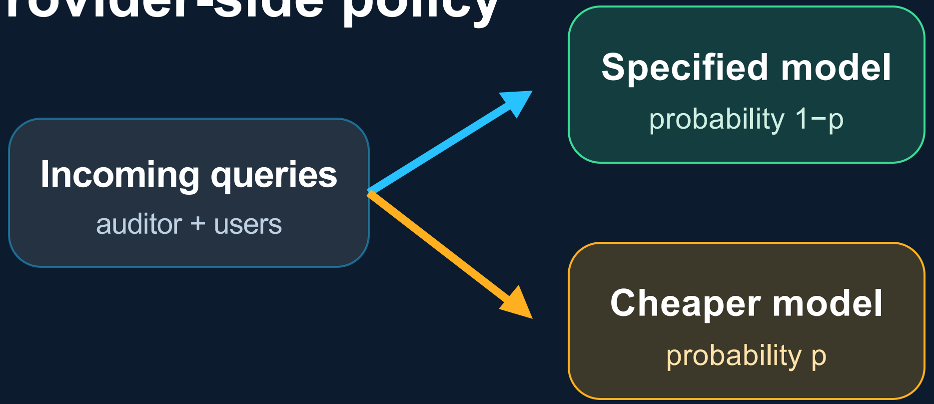
Mixture distribution

$$P_{mix} = (1 - p) \cdot P_{spec} + p \cdot P_{sub}$$

As p approaches 0, the mixed response distribution converges to the specified model distribution, shrinking the detectable signal.



Provider-side policy



Low-rate substitution attacks the sample budget

the brief characterizes detection as query-prohibitive

FAILURE MODE MATRIX

Software-Only Auditing Methods Are Fundamentally Unreliable

Method type	Weakness	Practical limitation
Text-based statistical tests (e.g., MMD)	Need many samples; subtle FP8 substitutions produce small distribution shifts.	Impractically large query budgets; randomized routing further reduces observable signal.
Log-probability methods	Same model yields different token log probabilities across software versions and GPU hardware.	Cannot distinguish legitimate infrastructure nondeterminism from actual substitution.
Adversarial countermeasures	Provider can detect known benchmarks or route only a fraction p of traffic to substitutes.	Benchmarks become unrepresentative; as $p \to 0$, detection becomes query-prohibitive.

Conclusion: software tests estimate behavior; they do not verify execution.

ACTIONABLE ALTERNATIVE

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

TEE attestation workflow

1. **Load model weights**
   inside hardware-secured enclave
2. **Run inference code**
   with protected execution state
3. **Produce attestation**
   cryptographic evidence of code + data
4. **Client verifies**
   correct model before trusting output

Why hardware attestation wins

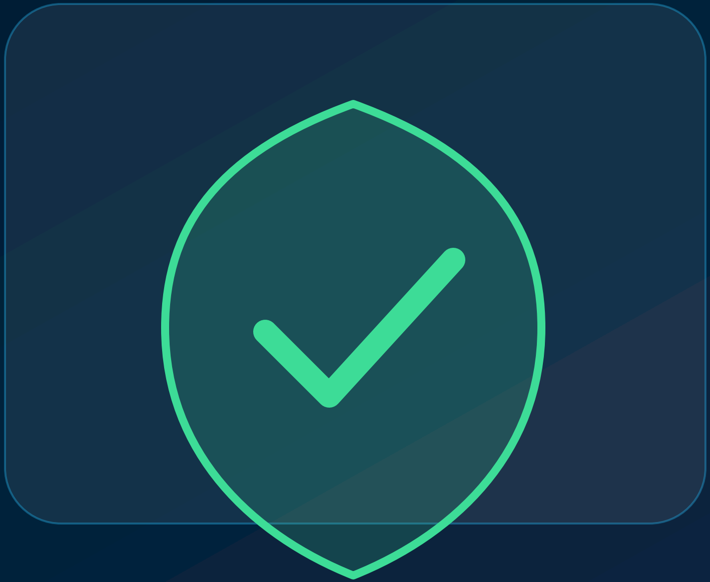
- **Provable**
  cryptographic attestation replaces statistical inference
- **Efficient**
  brief reports only modest performance overhead
- **Robust**
  not defeated by benchmark evasion or low-rate routing
- **Deployable**
  NVIDIA confidential computing on H100 GPUs is a practical path

Verify execution, not output statistics

KEY TAKEAWAYS

Close the Verification Gap with Hardware Attestation

1. **Substitution is economically motivated**
   black-box APIs let providers swap cheaper models while preserving premium pricing.
2. **FP8 barely moves many benchmarks**
   small score changes often sit within error bars, especially on MMLU and GSM8K.
3. **Log-prob fingerprints are unstable**
   framework and hardware nondeterminism defeat metadata-based comparison.
4. **Randomized routing attacks sample complexity**
   low p makes the mixed distribution converge toward the specified model.
5. **TEEs are the viable path today**
   hardware-secured attestation proves weights and inference code actually ran.



Community action

Make verifiable inference a cloud API requirement